

## Seoul Living Migration (Saenghwal Idong)

dong  $\times$  weekday  $\times$  arrival hour  $\times$  sex  $\times$  age band  $\times$  trip type

202001–202607 · 79 of 79 months · 424 dong

1 896 parquet files · 10186 891 962 rows

### Not recorded

- no date column, one row per weekday
- monthly sums over like weekdays, never daily means
- 26.1% of cells masked, nearly all of it in 20–44
- age-varying expansion weights, unrecorded
- a code space frozen at 2020, districts since moved

## External anchors

- official district file: 2.27 per masked cell, 11.1% of volume
- contact survey: pins  $\beta$  without identifying it
- resident registration: denominators, coverage profile

☐ data source

☐ estimation

☐ claim

☒ what the release does not record

## Co-arrival matrix

same dong, same arrival hour, same weekday  $\rightarrow$  an age  $\times$  age matrix A

at three spatial scales: 424 dong · 25 districts · 1 city

## Excess over proportionate mixing

assortativity · normalised mutual information · rank

### 1 The instrument responds

it tracks a school calendar it never sees

17 clearing months, all in term;  
none of 11 alternative calendars  
match

Fig 2

### 2 It still reads near zero

at every spatial and age resolution  
an identity bounds what it can read  
at all

Fig 3, Fig 4, Fig 5

### 3 The ranking moves

across the coverage family  
Kendall tau of the leading  
eigenvector falls to 0.117 (Fig 6);  
separately, 5 of 20 allocation  
verdicts clear their own null, worst  
at  $R_0$  1.2–1.5 (Fig S1)

Fig 6, Fig S1